

Figure 9.1 Heterogeneous military drone swarm architecture. ⏎

9.8.2 Loyal Wingman UAV – Human–machine teaming in air combat

The Loyal Wingman concept embodies a shift from isolated unmanned systems toward tightly coupled manned–unmanned teaming (MUM-T) in air combat, where autonomous or semi-autonomous UAVs operate alongside crewed aircraft as cooperative partners rather than as remote tools. A loyal wingman acts on high-level intent issued by a human pilot or mission commander and translates that intent into tactical actions: conducting forward reconnaissance, carrying sensors or effectors, serving as electronic-warfare nodes, absorbing enemy fire as expendable assets, or extending the sensor/ weapon reach of a manned platform. This model retains human responsibility for strategic decision-making while delegating hazardous, routine, or highly distributed tasks to the unmanned partner.

Technically, loyal wingman platforms combine robust autonomy, secure low-latency communications, redundant flight control, and modular payload architectures. They must be able to maintain formation with manned